TimeTensors forecasting benchmark

Executive summary of archived evidence

Decision. The fetched pre-schema commit contains a partial full constants-family run, not two completed experiment families. It preserves 132 completed seed markers and criterion curves, but no synchronized metric payload from which a full policy ranking can be rebuilt. No sampling-family artifact is present in commit `99b4d80`.

Imported full-run coverage

Dataset	Completed seed runs	Expected	Complete method/settings
ETTh1	45	45	15 / 15
Electricity	45	45	15 / 15
Traffic	42	45	14 / 15
Solar	0	45	0 / 15
Weather	0	45	0 / 15
Exchange Rate	0	45	0 / 15
Total	**132**	**270**	**44 / 90**

Table 1: Coverage of the intended constants grid: six datasets, three (L, H) settings, five policies, and three seeds. Job 42527 stopped with an out-of-memory error while starting Traffic 504:168 `drop_all_users`, seed 1.

Archived object	Count	Supports ranking?
`run.complete` signatures	132	completion only
Criterion-curve PDFs	132	optimization diagnostic only
`all_losses.pt` / `results.json` metrics	0	no
Full constants result tables	0	no
Fetched Slurm log pair	2 files	failure/progress audit
Sampling-family files	0	no

Table 2: The Git artifact policy excluded the tensor metrics required by the legacy table builder. Consequently the partial full run cannot answer which constant-handling policy is best, even for its 44 complete configurations.

Completed cluster smoke result

Electricity 504:168**, seed 1, 20 steps**	Test MSE	Constant windows
Keep all windows	1.16	0
Remove all constant windows	1.16	0
Drop all affected users	1.16	0

Table 3: Job 42510 validates the workflow but cannot distinguish policies: all 312 retained users have zero constant train, validation, and test windows.

Synthetic diagnostic retained before deletion

Family	Best observed MSE	Comparator	Measured toy result
Sampling	23.69	individual, batch 64	best sampled configuration
Linear models	23.39	least squares/ridge	beats trained linear Adam
Central/per-user	24.07	25.89	central pooling lower by 1.82
Losses	27.06	MSE	best toy loss

Table 4: Generator-specific 16-user, 360-date, $L = 24$, $H = 6$ smoke with two seeds and six epochs. These values validate implementation branches and are not PatchTST or real-dataset evidence.

What is known and unknown

Question	Quantitative answer
Did the full constants family complete?	No: 132/270 seed runs and 44/90 method/settings completed (Table 1).
Can constants policies be ranked?	No: zero synchronized metric payloads and zero full tables (Table 2).
Did the sampling family arrive?	No: zero sampling files in fetched commit `99b4d80` (Table 2).
Did the test distinguish policies?	No: all three methods have MSE 1.16 and zero constant windows (cluster-smoke table).
What survives scientifically?	Only the coverage/failure audit and the toy diagnostics quantified above; the real constants comparison must be rerun.

Archive boundary. The exact remote tree and log pair are stored under `outputs/archive/legacy_pre_schema_v1_2026-08-07/cluster_full_99b4d80`. They are excluded from current manifest readers and cannot conflict with the schema-1 `outputs/constants` hierarchy.

Source: archived commit 99b4d80, its constants tree, job 42527 logs, the earlier job 42510 table, and the archived synthetic smoke summary. Counts refer to non-empty files in the imported archive.